DEEP LEARNING

CORE CONCEPTS HANDBOOK

Computer Vision

Computer Vision is a field of Artificial Intelligence where machines learn to understand and analyze images or videos.

Humans can look at an image and instantly identify objects, faces, roads, animals, or emotions. Computer Vision tries to give similar ability to machines.

Used in:

- Face unlock in mobile phones
- CCTV monitoring
- Self-driving cars
- Medical image analysis
- Product quality inspection
- OCR text reading

Digital Image Structure

A digital image is made of many tiny squares called pixels.

If an image has size:

- 100 × 100 = 10,000 pixels
- 1920 × 1080 = 20,73,600 pixels

Each pixel stores color information.

The more pixels, the better image detail.

Pixel Values

A pixel is the smallest unit of an image.

For grayscale image:

- 0 = black
- 255 = white
- Between values = shades of gray
  $0 \leq$ Pixel Value ≤ 255

For color image, each pixel stores multiple values.

Grayscale and RGB Images

Grayscale Image

Contains only brightness values.

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Shape:

Height × Width × 1

Example:

28 × 28 × 1

RGB Image

Contains Red, Green, Blue channels.

Shape:

Height × Width × 3

Example:

224 × 224 × 3

Each pixel example:

- Red = 255
- Green = 120
- Blue = 50

Combining these gives final color.

Power of CNN :

Normal neural networks struggle with images because image size is huge.

Example:

224 × 224 × 3 = 150,528 inputs

Fully connecting every pixel creates too many parameters.

CNN solves this using:

- Shared filters
- Local pattern learning
- Feature extraction
- Reduced parameters

CNN is excellent for image tasks.

Convolution Layer

Convolution layer is the main learning layer of CNN.

It scans image using a small filter and detects patterns like:

- Edges
- Corners
- Shapes
- Texture

Instead of learning full image directly, CNN learns useful local patterns.

Filter / Kernel

A filter is a small matrix moving across image.

Example 3×3 filter:

Example 3×3 filter:

[

1	0	-1
1	0	-1
1	0	-1

]

This can detect vertical edges.

Different filters learn:

- Horizontal lines
- Curves
- Texture
- Complex shapes

Filters are learned during training.

Feature Map

After applying filter, output image is called feature map.

Feature map highlights where useful patterns are found.

If edge exists strongly, values become high in that region.

One filter creates one feature map. Many filters create many feature maps.

Stride

Stride means how many steps filter moves each time.

Stride = 1

Moves one pixel at a time.

More detail, larger output.

Stride = 2

Moves two pixels at a time.

Smaller output, faster computation.

Higher stride reduces output size.

Padding

Padding means adding extra border pixels around image.

Usually zeros are added.

Purpose:

- Keep border information
- Allow more filter movement

Without padding, image becomes smaller after convolution.

Activation in CNN

After convolution, activation function is applied.

Most common:

ReLU(x)=max(0,x)

Benefits:

- Removes negative values
- Adds non-linearity
- Faster learning

Used after almost every convolution layer.

Pooling Layer

Pooling reduces feature map size while keeping important information.

Benefits:

- Faster training
- Less memory usage
- Reduces overfitting
- Keeps strong features

Two common types:

- Max Pooling
- Average Pooling

Max Pooling

Selects maximum value from small region.

Example:

Example:

[

2	8
1	4

]

= 8

Average Pooling

Takes average of region.

Example:

[2 8

1 5]. →4

Used when smoother representation needed.

Flatten Layer

After several convolution and pooling layers, output is multi-dimensional.

Flatten converts it into one long vector.

Example:

7 × 7 × 64 becomes:

3136 values

This allows connection to dense layers.

Fully Connected Layer

This layer works like regular neural network.

It uses extracted features to make decision.

Example:

Detected whiskers + ears + eyes → Cat

Detected crack lines + rough surface → Surface Crack

Output Layer

Final layer gives prediction.

Binary Classification

One neuron + sigmoid

Crack / No Crack

Multi Class Classification

Multiple neurons + softmax

Cat / Dog / Horse

CNN Complete Flow

Image → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Flatten → Dense Layer → Output

This full pipeline converts raw image into prediction.

Training CNN Model

During training:

1. Give labeled images
2. CNN predicts output
3. Compare with actual label
4. Calculate loss
5. Update filters using backpropagation
6. Repeat many epochs

CNN gradually learns better filters automatically.

Image Classification

Image classification means assigning one label to full image.

Examples:

- Cat
- Dog
- Car
- Flower

Medical examples:

- Normal X-ray
- Pneumonia X-ray

Object Detection

Object detection means:

- Identify object type
- Locate object using bounding box

Examples:

- Detect person
- Detect vehicle
- Detect helmet

Popular models:

- YOLO

- SSD
- Faster R-CNN

Face Recognition

Steps:

- Detect face
- Extract features
- Compare with database

Uses:

- Attendance systems
- Security systems
- Mobile unlock

Medical Imaging

CNN is widely used in healthcare.

Examples:

- Tumor detection
- Pneumonia detection
- Fracture detection
- Diabetic retinopathy

Benefits:

- Fast screening
- Supports doctors
- Reduces manual effort

Industrial Inspection

Factories use CNN for defect detection.

Examples:

- Surface cracks
- Missing components
- Broken bottles
- PCB defects

Useful for automation and quality control.